

DO WE NEED COMPLEX TOPOLOGY CONTROL? DISTINCT-PEER RANDOM ROUTING IMPROVES COST-EFFICIENCY IN SPARSE MULTI-AGENT DEBATE

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ABSTRACT

Multi-agent debate (MAD) has emerged as a promising paradigm for improving the reasoning accuracy of large language models (LLMs) through iterative peer interaction. Communication topology plays a central role in this process, motivating increasingly sophisticated mechanisms that learn, adapt, or dynamically reconfigure agent interactions to improve accuracy or reasoning reliability. Meanwhile, prior studies suggest that much simpler sparse communication can already achieve competitive performance at substantially lower cost. In this work, we take a closer look at sparse MAD and ask whether complex topology control is actually necessary to improve collective reasoning. We find that a simple random-without-replacement routing policy, which lets each agent debate with two distinct and newly sampled peers at every round, provides a surprisingly strong baseline and consistently improves the accuracy-cost trade-off of sparse MAD. Building on this observation, we further study deliberation stopping and show that lightweight stopping can substantially reduce inference cost while preserving competitive accuracy. Our results suggest that sophisticated topology control such as learned topology adaption should be evaluated against strong simple routing and stopping baselines before its additional complexity is justified.

Index Terms— Multi-agent systems, Large language models, Multi-agent debate, Communication topology

1. INTRODUCTION

Multi-agent debate (MAD) is widely regarded as a paradigm that can improve reasoning quality of large language models (LLMs) through proposal, critique, and revision among agents [1–4]. Its communication topology determines whose intermediate solutions each agent can inspect, shaping both opportunities for correction and the spread of redundant or erroneous information. Prior work shows that sparse communication can match or sometimes exceed fully connected debate in answer accuracy using fewer tokens [5, 6].

Recent topology-control methods learn sparse masks, generate task-adaptive graphs, select interaction structures, or use graph diffusion [7–11]. The motivation is intuitive: better peer selection may retain useful information while avoiding redundant or misleading messages. Yet a simple neighbor-connected topology already offers competitive answer accuracy at lower token cost than dense debate [5]. Analyses of homogeneous debate also find that repeated interaction need not outperform majority voting over initial responses [12]. We therefore ask: *does additional topology control improve accuracy or reduce token cost beyond simple sparse routing and short debate?*

We study this question through a **two-axis analysis framework** (Fig. 1). Routing determines *who communicates*, while stopping determines *how long communication continues*. We compare reference topologies, schedule-based and uncertainty-guided routing,

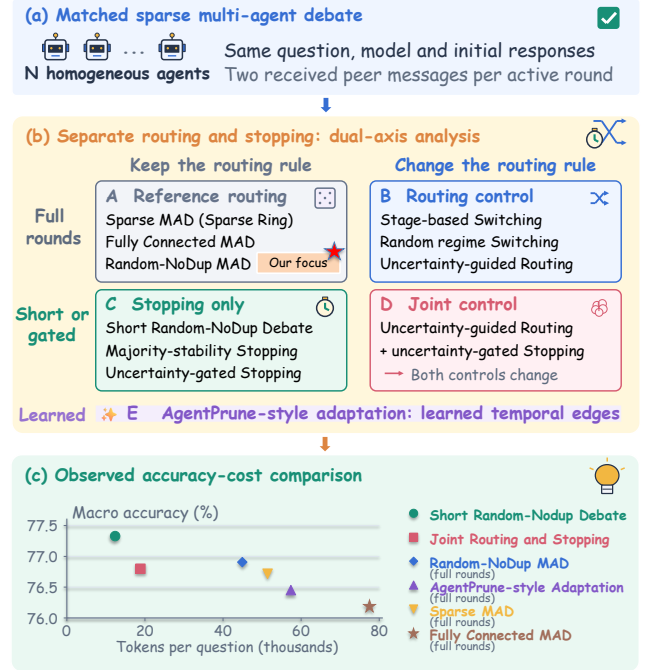


Fig. 1. An overview of the study. (a) Homogeneous sparse debate as the starting point. (b) The dual-axis routing and stopping evaluation framework. (c) Observed accuracy-cost comparison.

stopping-only policies, and their combination. A matched-protocol temporal-edge adaptation of AgentPrune provides a learned comparator [7]. This design separates the effect of choosing peers from the effect of reducing the number of rounds.

We focus on **Random-NoDup**, which samples distinct peers in each round and allows later revisits. We measure final-answer accuracy and total input plus output tokens per question. At full rounds, Random-NoDup has higher observed macro accuracy than Ring at similar token cost. Its short version has the highest observed macro accuracy and lowest mean token use among the tested methods.

In this paper, we primarily ask two research questions: **RQ1**, how competitive are simple routing baselines compared with structured, adaptive, and learned topology control? **RQ2**, does additional stopping control further improve the accuracy–cost trade-off? Our contributions are: (1) we introduce Random-NoDup as a simple distinct-peer sparse routing baseline; (2) we propose a two-axis evaluation framework for MAD, and empirically showing that Random-NoDup is highly competitive and stopping further improves its efficiency.

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2. RELATED WORK

Sparse multi-agent debate. Early MAD lets multiple model instances propose and revise solutions through repeated peer exchange [1, 13]. Li et al. systematically vary communication connectivity and show that sparse MAD can match or outperform fully connected debate while reducing computation [5]. Later work makes sparse communication more selective: CortexDebate constructs a sparse debating graph in which agents interact with peers estimated to be useful [14], while AgentPrune removes redundant edges from the spatial-temporal message-passing graph [7]. Shen et al. study information propagation directly, finding that moderate sparsity can suppress error diffusion while preserving beneficial information flow, and build EIB-Learner from this observation [15]. Complementarily, recent analysis of homogeneous MAD shows that additional debate is not automatically useful and that lightweight diversity or confidence interventions can outperform vanilla deliberation [12, 16]. These results motivate sparse and economical communication, but leave open how strong a simple random sparse baseline can be when peer collisions and deliberation length are controlled.

Topology control in MAD. A parallel line treats the communication graph itself as an optimization variable. GPTSwarm optimizes agent computational graphs [17, 18]; G-Designer generates task-adaptive topologies [8, 19]; and AMAS dynamically selects graph structures [9]. More recent work further learns communication structure through guided diffusion [10, 20] or reinforcement learning [21]. Small-world priors provide another structured alternative for balancing local and long-range communication [22]. Our focus is complementary: rather than designing a more expressive router, we ask whether topology control provides a clear advantage over strong simple routing under a matched sparse MAD protocol. We therefore compare against Sparse MAD and an AgentPrune-style learned temporal-edge adaptation, while evaluating the effect of stopping.

3. PRELIMINARIES AND METHODS

3.1. Sparse debate and reference topologies

We use n homogeneous agents $\mathcal{V}$ and at most T answer rounds. Agents answer independently at $t = 1$; later updates read k peers' responses from the previous round. Let $a_i^{(t)}$ denote the extracted answer of agent i at round t , and let $\mathcal{Y}_t$ be the set of distinct answers produced by all agents at that round. For each $y \in \mathcal{Y}_t$, we define the empirical answer distribution

$$\begin{aligned} p_t(y) &= \frac{1}{n} \sum_i \mathbf{1}[a_i^{(t)} = y], \quad m_t \in \arg \max_{y \in \mathcal{Y}_t} p_t(y), \\ C_t &= p_t(m_t), \quad H_t = - \sum_{y \in \mathcal{Y}_t} p_t(y) \log p_t(y). \end{aligned} \quad (1)$$

Here, $p_t(y)$ is the fraction of agents predicting answer y , m_t is the modal (majority) answer, C_t is the fraction of agents supporting that answer, and H_t measures disagreement among agent answers. In the following, we provide the descriptions and illustrations (as shown in Fig. 2) of our reference baselines:

Ring / Sparse MAD. Each receiver keeps the same local peers across rounds (Fig. 2(a)), yielding stable local information flow. This instantiates the neighbor-connected topology under our protocol [5].

Small-world-inspired routing. Each receiver combines persistent local communication with a resampled non-local shortcut (Fig. 2(b)). The local connection preserves continuity across rounds, while the shortcut exposes the receiver to a changing source of information.

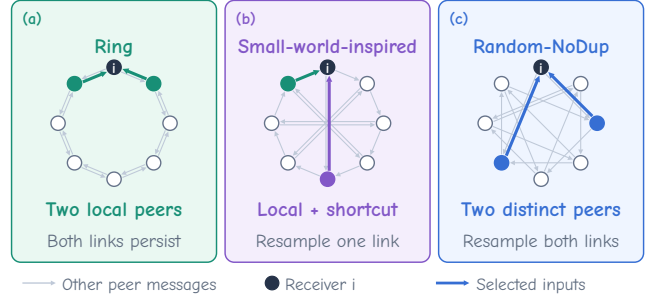


Fig. 2. Three reference routing rules with matched received degree. The example graphs share node positions; colored arrows highlight two inputs to receiver i . (a) Ring keeps both local peers. (b) Small-world-inspired routing keeps one local peer and resamples one non-local shortcut. (c) Random-NoDup resamples two distinct peers. Previously contacted peers remain eligible.

Random-NoDup routing. Each receiver draws k distinct peers at every active round (Fig. 2(c)):

$$\mathcal{N}_i^{(t)} \sim \text{Unif}\{S \subseteq \mathcal{V} \setminus \{i\} : |S| = k\}. \quad (2)$$

Earlier peers remain eligible. The graph varies across rounds, but the rule is state-agnostic. Distinct peers need not provide independent information.

3.2. Routing and stopping controls

Stage-based switching follows a preset schedule, while random switching selects a routing regime independently of agent states. Uncertainty-guided routing uses

$$U_t = \frac{H_t}{2 \log n} + \frac{1}{2n} \sum_{i=1}^n \mathbf{1}[a_i^{(t)} \neq a_i^{(t-1)}], \quad t \geq 2, \quad (3)$$

where H_t is answer entropy, n is the number of agents, and $a_i^{(t)}$ is agent i 's answer at round t . Thresholds $\theta_{\text{low}} < \theta_{\text{high}}$ map low, intermediate, and high U_t to Ring, small-world-inspired, and Random-NoDup routing. Thus U_t captures disagreement and answer changes rather than correctness.

Short Random-NoDup returns the majority answer m_h at a fixed horizon $h < T$. Adaptive stopping uses

$$\begin{aligned} g_t &= \mathbf{1}[m_t = m_{t-1}] \mathbf{1}[U_t < \theta_{\text{stop}}], \\ \tau &= \min \left(\left\{ t : \prod_{j=0}^{p-1} g_{t-j} = 1 \right\} \cup \{T\} \right), \end{aligned} \quad (4)$$

where θ_{stop} is the uncertainty threshold, p is the required number of stable transitions, and τ is the stopping round. The parameters are selected using held-out calibration sets. Removing the uncertainty condition gives majority-stability stopping, while joint control combines uncertainty-guided routing with the same stopping rule.

3.3. Learned temporal-edge adaptation

We also include an AgentPrune-style learned topology baseline [7]. AgentPrune assigns learnable probabilities to candidate communication edges, updates them from task rewards, and prunes low-value

Table 1. Five-benchmark results. Accuracy (%) and tokens (k/question) are mean $\pm$ std over five seeds; macro values are mean-only. Bold marks the best observed mean for each benchmark, not significance. **Red** numbers denote the best macro accuracy or token cost across methods.

ID	Method	Metric	ARC-Challenge	ScienceQA-text	GSM8K	MMLU-Pro	GPQA-Diamond	Macro
A Reference routing and prior sparse topology								
A1	Ring / Sparse MAD	Acc. $\uparrow$	90.67 $\pm$ 1.15	96.00 $\pm$ 0.58	90.67 $\pm$ 3.06	63.33 $\pm$ 3.06	40.67 $\pm$ 1.15	76.27
		Tok. $\downarrow$	35.99 $\pm$ 0.35	33.32 $\pm$ 0.11	67.76 $\pm$ 0.10	64.74 $\pm$ 0.14	84.43 $\pm$ 0.58	57.25
A2	Small-world-inspired	Acc. $\uparrow$	90.67 $\pm$ 1.15	95.33 $\pm$ 1.15	92.67 $\pm$ 1.15	64.67 $\pm$ 2.31	42.00 $\pm$ 0.00	77.07
		Tok. $\downarrow$	35.90 $\pm$ 0.09	33.15 $\pm$ 0.20	68.00 $\pm$ 0.10	65.10 $\pm$ 0.40	85.46 $\pm$ 0.51	57.52
A3	Random-NoDup, full deliberation	Acc. $\uparrow$	91.33 $\pm$ 1.15	95.67 $\pm$ 1.15	90.00 $\pm$ 3.46	65.33 $\pm$ 3.06	42.00 $\pm$ 3.46	76.87
		Tok. $\downarrow$	35.69 $\pm$ 0.04	33.17 $\pm$ 0.11	67.95 $\pm$ 0.18	65.04 $\pm$ 0.28	85.04 $\pm$ 0.50	57.38
B Routing control with full deliberation								
B1	Stage-based topology switching	Acc. $\uparrow$	90.00 $\pm$ 2.00	96.00 $\pm$ 2.00	88.00 $\pm$ 2.00	63.33 $\pm$ 1.15	41.33 $\pm$ 4.16	75.73
		Tok. $\downarrow$	35.79 $\pm$ 0.15	33.29 $\pm$ 0.18	67.70 $\pm$ 0.15	65.10 $\pm$ 0.20	85.07 $\pm$ 1.13	57.39
B2	Random topology switching	Acc. $\uparrow$	91.33 $\pm$ 1.15	94.00 $\pm$ 2.00	91.33 $\pm$ 1.15	63.33 $\pm$ 4.16	39.33 $\pm$ 4.16	75.86
		Tok. $\downarrow$	35.70 $\pm$ 0.14	33.27 $\pm$ 0.09	68.00 $\pm$ 0.10	65.10 $\pm$ 0.60	85.43 $\pm$ 0.78	57.50
B3	Uncertainty-guided routing	Acc. $\uparrow$	90.67 $\pm$ 1.15	94.00 $\pm$ 1.39	93.33 $\pm$ 1.15	64.00 $\pm$ 2.00	40.67 $\pm$ 3.06	76.53
		Tok. $\downarrow$	35.78 $\pm$ 0.23	33.23 $\pm$ 0.11	67.80 $\pm$ 0.20	65.00 $\pm$ 0.20	84.81 $\pm$ 1.02	57.32
C Random-NoDup with stopping control								
C1	Random-NoDup, fixed round 3	Acc. $\uparrow$	92.00 $\pm$ 3.46	94.00 $\pm$ 2.00	92.00 $\pm$ 0.00	65.33 $\pm$ 1.15	43.33 $\pm$ 2.31	77.33
		Tok. $\downarrow$	8.05 $\pm$ 0.03	7.41 $\pm$ 0.03	13.11 $\pm$ 0.02	14.01 $\pm$ 0.07	19.20 $\pm$ 0.13	12.36
C2	Majority-stability stopping	Acc. $\uparrow$	92.00 $\pm$ 2.00	92.67 $\pm$ 1.15	90.67 $\pm$ 1.15	62.00 $\pm$ 0.00	43.33 $\pm$ 4.16	76.13
		Tok. $\downarrow$	8.13 $\pm$ 0.03	7.42 $\pm$ 0.01	13.80 $\pm$ 0.40	16.20 $\pm$ 0.60	23.55 $\pm$ 0.75	13.82
C3	Uncertainty-gated stopping	Acc. $\uparrow$	93.33 $\pm$ 1.15	94.00 $\pm$ 0.79	92.00 $\pm$ 0.00	62.67 $\pm$ 5.03	44.00 $\pm$ 5.29	77.20
		Tok. $\downarrow$	9.72 $\pm$ 0.64	7.75 $\pm$ 0.05	16.59 $\pm$ 1.82	20.66 $\pm$ 0.20	38.27 $\pm$ 2.35	18.60
D Joint routing and stopping control								
D1	Uncertainty-guided routing + stopping	Acc. $\uparrow$	92.67 $\pm$ 1.15	93.33 $\pm$ 1.15	92.00 $\pm$ 2.00	64.00 $\pm$ 0.00	42.00 $\pm$ 4.00	76.80
		Tok. $\downarrow$	9.66 $\pm$ 0.82	7.74 $\pm$ 0.07	17.60 $\pm$ 0.90	19.30 $\pm$ 1.80	39.90 $\pm$ 7.69	18.84
E Learned topology adaptation								
E1	AgentPrune-style temporal-edge adaptation	Acc. $\uparrow$	91.33 $\pm$ 1.15	94.00 $\pm$ 2.00	89.33 $\pm$ 3.06	66.67 $\pm$ 4.62	40.00 $\pm$ 5.29	76.27
		Tok. $\downarrow$	35.75 $\pm$ 0.20	33.39 $\pm$ 0.12	67.88 $\pm$ 0.20	65.21 $\pm$ 0.03	85.70 $\pm$ 1.13	57.59

edges to obtain a sparse topology. For a matched comparison, we adapt its learned edge-selection mechanism to our sparse MAD setting using disjoint calibration questions, while keeping the agents, prompts, synchronous updates, received degree, and aggregation rule unchanged. This provides a learned topology-control baseline under the same sparse debate protocol as the other methods.

4. EXPERIMENTS

4.1. Experimental Setup

Homogeneous agents and benchmarks. We study homogeneous MAD, where all eight agents in a debate use the same LLM. The main experiments use GPT-4o-mini on ARC-Challenge [31], ScienceQA-text [32], GSM8K [33], MMLU-Pro [34], and GPQA-Diamond [35]. ScienceQA-text contains text-only questions. across models.

Debate protocol. Each debate contains eight agents, and each agent receives messages from two peers per communication round. Round 1 consists of independent answers, followed by synchronous debate updates. Full debate runs for 14 rounds, while Short Random-NoDup stops at round 3 after two communication updates. All methods use the same generation settings (e.g., temperature and prompts) within each matched comparison. Results are reported as mean $\pm$ standard deviation over five independent runs with random seeds.

Routing and stopping controls. We compare the reference topologies with stage-based, random, and uncertainty-guided routing, as well as fixed and adaptive stopping. Stage-based routing switches after round 5. For uncertainty-based control, we use $\theta_{\text{low}} = \theta_{\text{stop}} = 1/3$, $\theta_{\text{high}} = 2/3$, and patience $p = 2$. The learned-edge baseline uses separate calibration questions. We evaluate each method by final-answer accuracy and inference token cost, with macro results averaged equally across tasks.

4.2. RQ1: Does Topology Control Improve Routing?

We test whether topology control improves accuracy beyond simple routing by comparing reference rules and controllers at the same degree and full horizon. Table 1 reports accuracy and token cost.

Reference routing. Ring ties for the highest ScienceQA-text accuracy, while small-world-inspired routing has the highest macro accuracy among the three reference rules. Random-NoDup exceeds Ring in mean accuracy on ARC-Challenge, MMLU-Pro, and GPQA-Diamond, and in macro accuracy, with similar token use. Figure 2 shows their key difference: fixed local peers, one refreshed shortcut, or two resampled distinct peers. No single pattern gives the highest accuracy on every task.

Additional routing control. Full-round Random-NoDup has higher macro accuracy than all three routing controllers and the learned-edge adaptation. Uncertainty-guided routing leads on GSM8K, and the learned adaptation leads on MMLU-Pro, but these gains do not carry across tasks. Full-round token costs remain close, and learned edges require calibration. Additional control can therefore help individual tasks without improving the overall trade-off.

Overall comparison. Across the matched full-round comparisons, Random-NoDup remains competitive with structured, adaptive, and learned routing while requiring no state estimation or topology optimization. Results show that additional topology control does not provide a consistent accuracy–cost advantage over this baseline. We next examine whether stopping offers a clearer efficiency gain.

4.3. RQ2: Does Stopping Improve the Trade-off?

We next keep Random-NoDup routing fixed and compare full rounds, a short fixed cutoff, and two answer-based stopping gates. This experiment evaluates whether implementing an additional stopping criterion can increase accuracy while reducing token consumption.

Table 2. Cross-model accuracy (%) and tokens (k/question): mean above $\pm$ SD. Bold: best mean per model-task pair. Round 3 is a same-trajectory cutoff; other runs use full rounds.

Method	ARC-Challenge		GSM8K		GPQA-Diamond	
	Acc. $\uparrow$	Tok. $\downarrow$	Acc. $\uparrow$	Tok. $\downarrow$	Acc. $\uparrow$	Tok. $\downarrow$
GPT-4o-mini						
Ring /	90.67	35.99	90.67	67.76	40.67	84.43
Sparse MAD	± 1.15	± 0.35	± 3.06	± 0.10	± 1.15	± 0.58
Random-NoDup	91.33	35.69	90.00	67.95	42.00	85.04
(full rounds)	± 1.15	± 0.04	± 3.46	± 0.18	± 3.46	± 0.50
Random-NoDup	92.67	8.04	90.00	13.12	48.67	19.26
(round 3)	± 2.31	± 0.00	± 0.00	± 0.03	± 5.03	± 0.12
Uncertainty-guided	90.67	35.78	93.33	67.78	40.67	84.81
routing	± 1.15	± 0.23	± 1.15	± 0.02	± 3.06	± 1.02
GPT-4.1-mini						
Ring /	94.67	35.30	94.00	53.38	62.67	96.69
Sparse MAD	± 1.15	± 0.16	± 2.00	± 0.32	± 1.15	± 0.14
Random-NoDup	94.67	35.31	94.00	53.50	66.00	98.08
(full rounds)	± 1.15	± 0.16	± 0.00	± 0.72	± 2.00	± 0.34
Random-NoDup	94.00	8.18	94.67	10.56	66.67	23.53
(round 3)	± 2.00	± 0.01	± 1.15	± 0.02	± 3.06	± 0.15
Uncertainty-guided	92.67	35.43	94.00	53.65	63.33	97.58
routing	± 1.15	± 0.12	± 2.00	± 0.66	± 1.15	± 0.44
GPT-4.1						
Ring /	92.67	27.89	96.67	40.00	62.67	54.13
Sparse MAD	± 1.15	± 0.02	± 1.15	± 0.10	± 1.15	± 0.48
Random-NoDup	93.33	27.88	96.00	40.13	62.67	53.89
(full rounds)	± 1.15	± 0.02	± 2.00	± 0.03	± 1.15	± 0.26
Random-NoDup	94.67	7.02	91.33	9.06	60.67	17.38
(round 3)	± 1.15	± 0.01	± 1.15	± 0.04	± 1.15	± 0.16
Uncertainty-guided	92.00	27.88	97.33	40.19	62.67	54.10
routing	± 0.00	± 0.01	± 1.15	± 0.18	± 3.06	± 0.34
GPT-5.6 Luna						
Ring /	84.67	26.86	98.00	26.80	54.00	46.77
Sparse MAD	± 1.15	± 0.01	± 0.00	± 0.01	± 3.46	± 0.02
Random-NoDup	84.67	26.86	98.67	26.80	54.67	46.78
(full rounds)	± 1.15	± 0.01	± 1.15	± 0.01	± 4.16	± 0.00
Random-NoDup	88.00	5.93	99.33	6.42	58.67	10.89
(round 3)	± 0.00	± 0.00	± 1.15	± 0.01	± 2.31	± 0.01
Uncertainty-guided	84.00	26.86	97.33	26.79	52.00	46.77
routing	± 0.00	± 0.01	± 2.31	± 0.01	± 2.00	± 0.03

Fixed and adaptive stopping. Short Random-NoDup achieves the best observed macro accuracy while using the fewest tokens (Table 1). Adaptive stopping can improve some individual tasks, but at higher cost, and adding routing control does not improve the overall trade-off. Paired results consistently favor the short setting in token cost, while accuracy differences are generally small.

Why use round 3? Figure 3(a) tests cutoffs of complete trajectories. Macro accuracy is close to its observed peak during rounds 2–5, while later rounds add tokens. Round 3 includes two communication updates, and its paired accuracy intervals against rounds 2, 4, and 5 include zero. It is an early reference, not a uniquely optimal cutoff. These prefixes are separate from the independent short runs in Table 1.

Overall accuracy–cost trade-off. Across the stopping variants, the main benefit of short debate is improved efficiency rather than a uniform accuracy gain. As shown in Table 1, the short Random-NoDup setting achieves accuracy comparable to the longer and adaptive alternatives while using substantially fewer tokens. Figure 3 further shows that most of the useful performance is already reached in the early rounds, whereas additional debate continues to increase cost without a consistent accuracy improvement. Thus, for RQ2, short debate

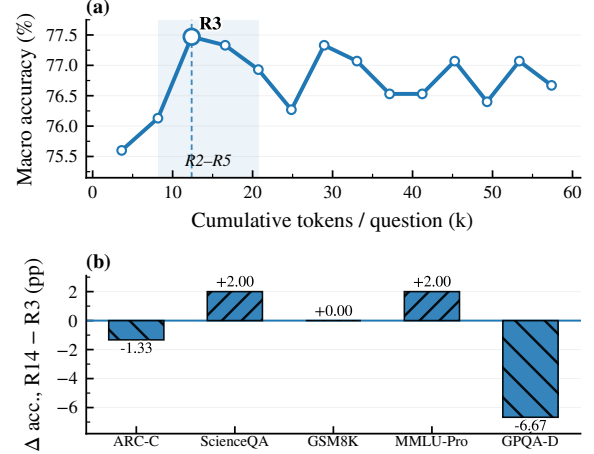


Fig. 3. Random-NoDup horizon analysis. (a) Accuracy versus token cost for cutoffs of the same trajectories; the shaded region covers rounds 2–5. (b) Per-task accuracy change from round 3 to round 14.

provides a stronger accuracy–cost trade-off by reducing inference cost while preserving competitive accuracy.

4.4. Ablations on Cross-model Performance

We repeat four key comparisons across four model variants to check whether the findings depend on the primary model. Table 2 compares Ring, full and short Random-NoDup, and uncertainty-guided routing on the same three tasks. Short results use the round-3 prefix of each matched full Random-NoDup trajectory.

Routing across models. At full rounds, neither Ring nor uncertainty-guided routing establishes a paired accuracy advantage over Random-NoDup in any of the twelve model–task comparisons. The intervals also do not establish consistent superiority for Random-NoDup. Thus, simple random routing remains a competitive reference across the tested variants, not only in the primary model.

Stopping across models. Short Random-NoDup uses the fewest tokens for every model–task pair and has the highest macro accuracy for three of the four models. However, one model benefits from longer debate on GSM8K, where stopping at round 3 reduces accuracy. The cost savings transfer more consistently than accuracy preservation. Overall, a short debate paradigm with distinct-peer routing constitutes a robust baseline that generalizes across the evaluated models.

5. CONCLUSION

In this paper, we study sparse multi-agent debate through a two-axis framework that separates peer routing from stopping. The comparisons cover reference topologies, switching policies, uncertainty-guided control, joint control, and a matched-protocol AgentPrune-style temporal-edge adaptation. Experimental results show that the proposed distinct-peer random routing method, Random-NoDup, provides competitive accuracy without learned or state-aware routing, while short debate further improves the token savings.

Cross-model results support this distinction: additional routing control shows no detected accuracy advantage over Random-NoDup in the tested comparisons, whereas stopping consistently reduces token use. Its accuracy effect still depends on the model and task. Our findings indicate that advanced topology control methods, including learned topology adaptation and switching methods should first be benchmarked against the proposed distinct-peer random routing baselines before their added complexity can be warranted.

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COMPLIANCE WITH ETHICAL STANDARDS

This work involves no human participants or newly collected human-subject data. All experiments rely on public benchmarks and model inference conducted under applicable terms of use.

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